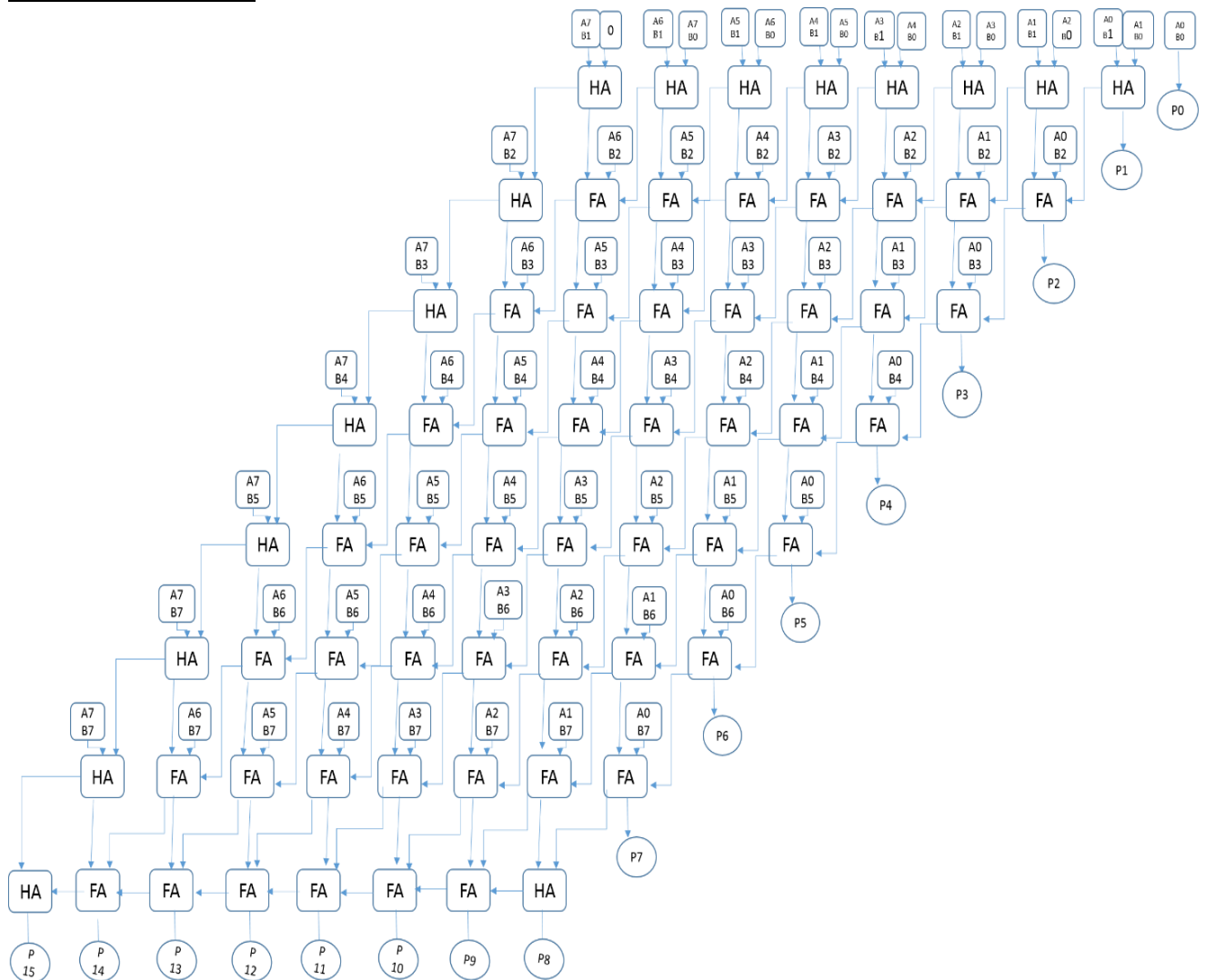


8-BIT CARRY SAVE MULTIPLIER

INTRODUCTION

Multipliers play an important role in digital signal processing and various other applications. Some of the design targets to be kept in mind are high speed, low power consumption, less area etc. Out of the many multiplier designs available we have chosen to build an 8-bit carry save multiplier using half adder blocks and full adder blocks. The layout is designed using MAGIC tool. The full adder and half adder blocks were built using NAND standard cells available on 'vsclib' folder.

CIRCUIT DIAGRAM



DESCRIPTION :

The circuit was built using Half Adder and Full Adder blocks and these adders are used for the accumulation of partial product. A carry-propagate adder is used for the generation of final product which reduces the critical path delay.

Two 8 – bit inputs are given and 16 partial products are obtained. Each partial product is generated by the multiplication of the multiplicand with one multiplier bit. The partial product are shifted according to their bit orders and then added. The final output is a 16 bit number.

				A3	A2	A1	A0	Inputs
		x		B3	B2	B1	B0	
			C	B0 x A3	B0 x A2	B0 x A1	B0 x A0	Internal Signals
		+	B1 x A3	B1 x A2	B1 x A1	B1 x A0		
		C	sum	sum	sum	sum		
	+	B2 x A3	B2 x A2	B2 x A1	B2 x A0			
	C	sum	sum	sum	sum			
+	B3 x A3	B3 x A2	B3 x A1	B3 x A0				
C	sum	sum	sum	sum				Outputs
Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	

SIMULATION & RESULTS

LAYOUT

OBSERVATION

<u>PARAMETER</u>	<u>VALUE</u>
Clock Frequency	
No of gates	430
Area	2504 x 1998
Capacitance	147.7 pF
Critical Path	

Power	
No of transistors	3431

APPLICATION

The type of multiplier we choose is based on application. For high speeds often parallel multipliers are used. Here the no of partial products is the factor that determines performance. Shift and Add algorithm is also most popular for multiplication. However with increasing no of partial products and shifted sums the performance speed can deteriorate. Hence we use a combination of serial-parallel multipliers to achieve better design targets like lower area and power consumption while compromising on high speed.

The carry save multiplier is a modification to the well-known array multiplier where the carry ripple adder has been replaced by carry save adders to improve on delay and area.

CONCLUSION